

Please check the examination details below before entering your candidate information

Candidate surname		Other names	
Centre Number		Candidate Number	

Pearson Edexcel Award

Tuesday 25 June 2024

Afternoon (Time: 3 hours)	Paper reference	9811/01
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Advanced Extension Award

Mathematics

You must have: Mathematical Formulae and Statistical Tables An insert for questions 5, 6 and 7 (enclosed)	Total Marks
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Instructions

- Use **black** ink or ball-point pen.
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- **Calculators may not be used.**
- You must **show all your working.**
- Answers should be given in as simple a form as possible, e.g. $\frac{2\pi}{3}$, $\sqrt{2}$, $3\sqrt{2}$.

Information

- The total mark for this paper is 100 of which **7** marks are for style and clarity of presentation.
- The style and clarity of presentation marks will be indicated as **(+S1) or (+S2)**.
- There are 7 questions in this question paper.
- The marks for **each** question are shown in brackets.
- The total mark for each question is shown at the end of the question.

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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1. In the binomial expansion of

$$(1 - 8x)^p \quad |x| < \frac{1}{8}$$

where p is a positive constant,

- the sum of the coefficient of x and the coefficient of x^2 is equal to the coefficient of x^3
- the coefficient of x^2 is positive

Determine the value of p .

(7)



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Question 1 continued

Lined area for writing the answer to Question 1.

(Total for Question 1 is 7 marks)



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Question 2 continued

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(Total for Question 2 is 6 marks)



3. (i) Determine the value of k such that

$$\arctan \frac{1}{2} - \arctan \frac{1}{3} = \arctan k \quad (3)$$

- (ii)(a) Prove that

$$\cos 3A \equiv 4\cos^3 A - 3\cos A \quad (3)$$

Given that $a = \cos 20^\circ$

- (b) write down, in terms of a , an expression for $\cos 40^\circ$ (1)

- (c) determine, in terms of a , a simplified expression for $\cos 80^\circ$ (2)

- (d) Use part (a) to show that

$$4a^3 - 3a = \frac{1}{2} \quad (1)$$

- (e) Hence, or otherwise, show that

$$\cos 20^\circ \cos 40^\circ \cos 80^\circ = \frac{1}{8} \quad (4)$$

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Question 3 continued

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Question 3 continued

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Question 3 continued

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(Total for Question 3 is 14 marks)



4. (a) Use the substitution $x = \sqrt{3} \tan u$ to show that

$$\int \frac{1}{3 + x^2} dx = p \arctan(px) + c$$

where p is a real constant to be determined and c is an arbitrary constant.

(5)

- (b) Use the substitution $x = \frac{3u + 3}{u - 3}$ to determine the exact value of I where

$$I = \int_{-3}^1 \frac{\ln(3 - x)}{3 + x^2} dx$$

giving your answer in simplest form.

(10)

(+S1)



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Question 4 continued

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Question 4 continued

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Question 4 continued

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(Total for Question 4 is 16 marks)



5.

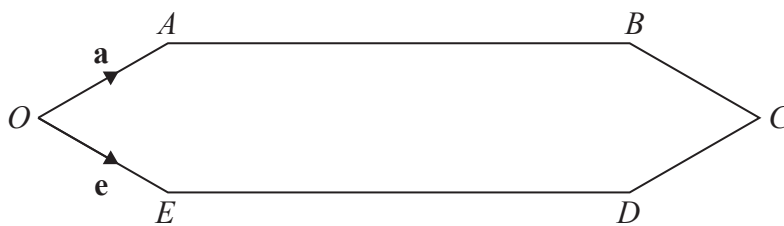


Figure 2

Figure 2 shows a sketch of a hexagon $OABCDE$ where

- the interior angle at O and at C are each 60°
- the interior angle at each of the other vertices is 150°
- $OA = OE = BC = CD$
- $AB = ED = 3 \times OA$

Given that $\vec{OA} = \mathbf{a}$ and $\vec{OE} = \mathbf{e}$

determine as simplified expressions in terms of $\mathbf{a}$ and $\mathbf{e}$

(a) $\vec{AB}$ (3)

(b) $\vec{OD}$ (2)

The point R divides AB internally in the ratio $1:2$

(c) Determine $\vec{RC}$ as a simplified expression in terms of $\mathbf{a}$ and $\mathbf{e}$ (2)

The line through the points R and C meets the line through the points O and D at the point X .

(d) Determine $\vec{OX}$ in the form $\lambda\mathbf{a} + \mu\mathbf{e}$, where λ and μ are real values in simplest form. (6)
(+S2)



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Question 5 continued

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Question 5 continued

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Question 5 continued

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Question 5 continued

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Question 5 continued

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(Total for Question 5 is 15 marks)



6.

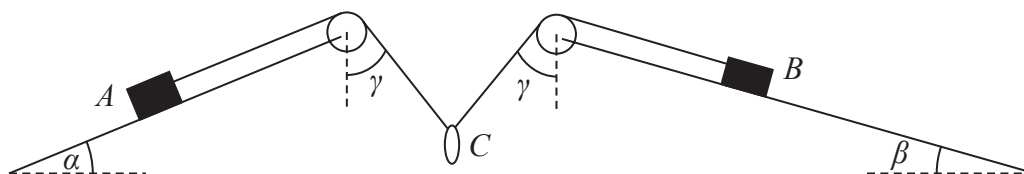


Figure 3

Figure 3 shows a block A with mass $4m$ and a block B with mass $5m$.
 Block A is at rest on a rough plane inclined at an angle α to the horizontal.
 Block B is at rest on a rough plane inclined at an angle β to the horizontal.

The blocks are connected by a light inextensible string which passes over a small smooth pulley at the top of each plane.

A small smooth ring C , of mass $8m$, is threaded on the string between the pulleys so that A , B and C all lie in the same vertical plane.

The part of the string between A and its pulley lies along a line of greatest slope of the plane of angle α .

The part of the string between B and its pulley lies along a line of greatest slope of the plane of angle β .

The angle between the vertical and the string between each pulley and the ring C is γ .

The two blocks, A and B , are modelled as particles.

Given that

- $\tan \alpha = \frac{5}{12}$ and $\tan \beta = \frac{7}{24}$ and $\tan \gamma = \frac{3}{4}$
- the coefficient of friction, μ , is the same between each block and its plane
- one of the blocks is on the point of sliding up its plane
- the tension in the string is T

(a) determine, in terms of m and g , an expression for T , (3)

(b) draw a diagram showing the forces on block A , clearly labelling each of the forces acting on the block, (3)

(c) determine the value of μ , giving a justification for your answer. (10)
(+S2)



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Question 6 continued

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Question 6 continued

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Question 6 continued

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Question 6 continued

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Question 6 continued

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(Total for Question 6 is 18 marks)



7.

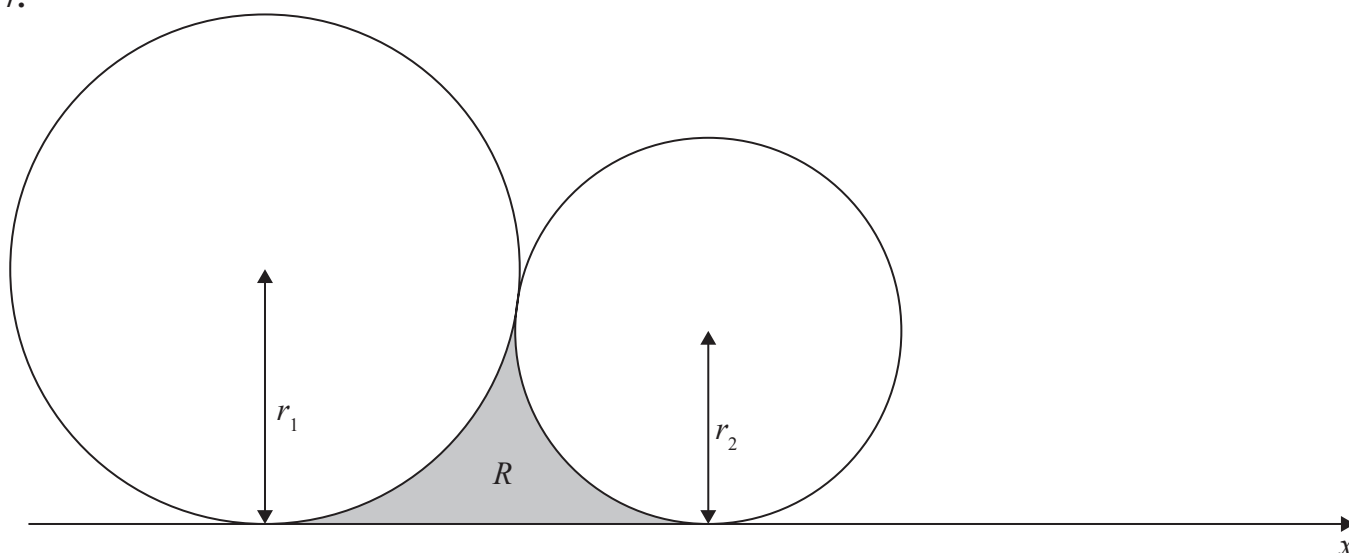


Figure 4

Figure 4 shows a circle with radius r_1 and a circle with radius r_2 . The circles touch externally at a single point above the x -axis. Both circles also have the x -axis as a tangent.

- (a) Show that the horizontal distance between the centres of the circles, d , is given by

$$d^2 = 4r_1 r_2 \quad (2)$$

The finite region R , shown shaded in Figure 4, is bounded by the x -axis and minor arcs of the two circles.

Given that $r_1 \geq r_2$

- (b) show that the area of R is given by

$$(r_1 + r_2)\sqrt{r_1 r_2} - \frac{1}{2}(r_1^2 - r_2^2)\theta - \frac{1}{2}\pi r_2^2$$

$$\text{where } \cos \theta = \frac{r_1 - r_2}{r_1 + r_2} \quad (5)$$

Question 7 continues on the next page.



Question 7 continued

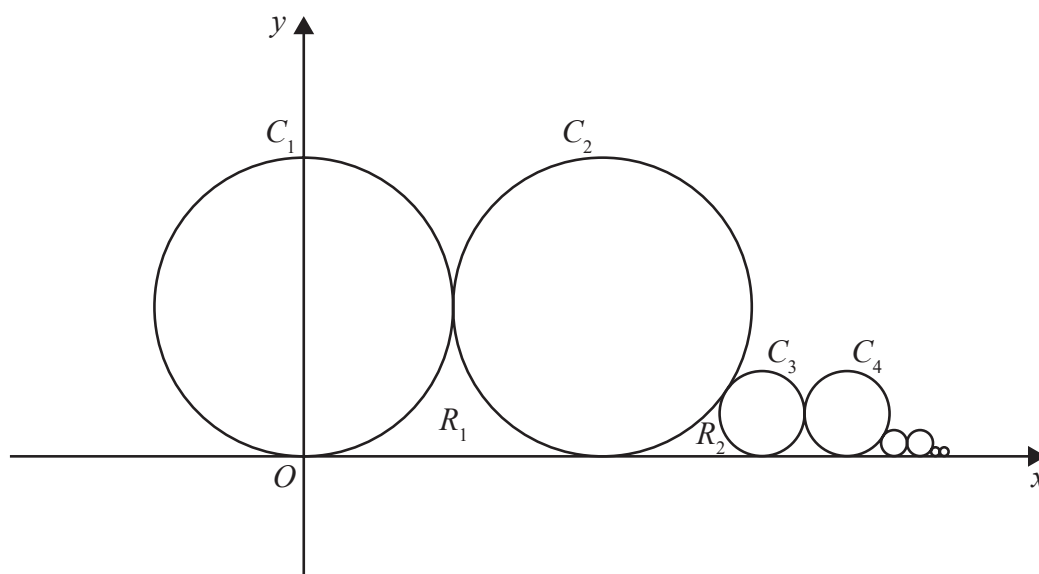


Figure 5

A sequence of circles, $C_1, C_2, C_3, \dots$ with radii $r_1, r_2, r_3, \dots$ respectively, is constructed such that

- each circle is tangential to and above the x -axis
- the first circle, C_1 , has centre $(0, 1)$
- each successive circle touches the preceding one externally at a single point
- the horizontal distances between the centres of successive circles form a geometric sequence with first term 2 and common ratio $\frac{1}{\sqrt{3}}$

The first few circles in the sequence are shown in Figure 5.

(c) (i) Determine the value of r_3

(ii) Show that, for $n \geq 1$, $r_{n+2} = kr_n$ where k is a constant to be determined.

(iii) Hence show that, for $n \geq 1$, $r_{2n} = r_{2n-1}$

(9)

The region bounded between C_n, C_{n+1} and the x -axis is R_n

The total area, A , bounded above the x -axis and under all the circles is the sum of the areas of all these regions.

(d) Determine the value of A , giving the answer in simplest form.

(6)

(+S2)

Question 7 continued

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Question 7 continued

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Question 7 continued

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Question 7 continued

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(Total for Question 7 is 24 marks)

TOTAL FOR PAPER IS 100 MARKS

